

Next-Generation Delta-Sigma Converters: Trends and Perspectives

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Abstract—This paper presents an overview of emerging circuits and systems techniques which are at the forefront of the state of the art in $\Delta\Sigma$ modulators, pushing their performance forward and giving rise to new generations of data converters. Among others, those strategies involving the development of new applications and paradigms—like RF/GHz-range $\Delta\Sigma$ digitisation, digital-assisted embedded loop filters, time-to-digital conversion and hybrid $\Delta\Sigma$ /Nyquist-rate architectures—are discussed, as well as the implications and design challenges derived from their integration in deep nanometer CMOS technologies. The envisioned $\Delta\Sigma$ techniques are presented in a systematic way around the main analog building blocks embedded in a $\Delta\Sigma$ modulator, i.e., the loop filter and the quantizer. Analysing the trends in the design of these blocks allows us to offer perspectives on how $\Delta\Sigma$ converters will evolve in the next years.

Index Terms—Delta-Sigma modulation, oversampling analog-to-digital conversion.

I. INTRODUCTION

THE PERVASIVE tendency towards digital devices and technologies in a data-driven world demands efficient ways of interacting with diverse physical environments by means of smart sensors, actuators, analog signal conditioning and analog-digital interfaces. In many cases, the latter (analog) blocks must be integrated into mainstream (digital) CMOS in order to reduce cost and size. Hence, they are forced to grapple with the analog impairments of nanoscale technologies, such as short-channel effects, low supply voltage and process variability [1], [2].

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Data converters are key enablers for the practical implementation of present digital technologies and future cyber-physical systems connected in the so-called Internet-of-Things (IoT). The state of the art on Analog-to-Digital Converters (ADCs) is currently dominated by three different techniques, namely: Pipeline, Successive Approximation Register (SAR) and Delta-Sigma Modulators ($\Delta\Sigma$ Ms), which year after year, continue to push the limits of energy efficiency [3]. Compared to other conversion techniques, $\Delta\Sigma$ ADCs cover the widest conversion region of the resolution-versus-bandwidth plane, providing the most efficient solution to digitise diverse types of signals in many different applications—from ultra-low-power biomedical devices to ultra-wide-band communications [4]. Moreover, $\Delta\Sigma$ ADCs are able to achieve high accuracy despite using inaccurate components by the combined action of oversampling and noise shaping. These advantages, together with their relaxed anti-aliasing requirements and their adaptability in many practical situations, are some of the reasons why many engineers nowadays, in both industry and academia, consider $\Delta\Sigma$ Ms as a first choice for their designs [5], [6].

The original concepts and fundamental signal processing techniques behind $\Delta\Sigma$ Ms have evolved a lot over the last few years, giving rise to many architectures, circuits and systems techniques. However, new application scenarios require looking for innovative solutions to meet increasingly aggressive specifications in a power efficient manner, thus leading to a new generation of $\Delta\Sigma$ converters. Examples include Radio-Frequency (RF) digitizers for Software-Defined Radios (SDRs), in which part of the RF signal processing is embedded in the $\Delta\Sigma$ loop [7]. Other trends involve the intensive use of digitally assisted analog circuits [8], [9] or time-to-digital conversion [10], thus taking advantage of digital signal processing integrated at the nanoscale.

This paper aims to provide a survey of some of the most promising techniques, giving future directions, challenges and visions in the design of next-generation $\Delta\Sigma$ Ms, as well as the implications of their integration in analog-hostile mainstream technologies. The rest of this paper is organized as follows. Section II presents an overview of the state of the art on $\Delta\Sigma$ ADCs, by comparing their performance metrics with Nyquist-Rate converters. Sections III and IV outline some cutting-edge $\Delta\Sigma$ Ms, which state new trends in the design of the $\Delta\Sigma$ loop-filter (Section III) and quantizer (Section IV).

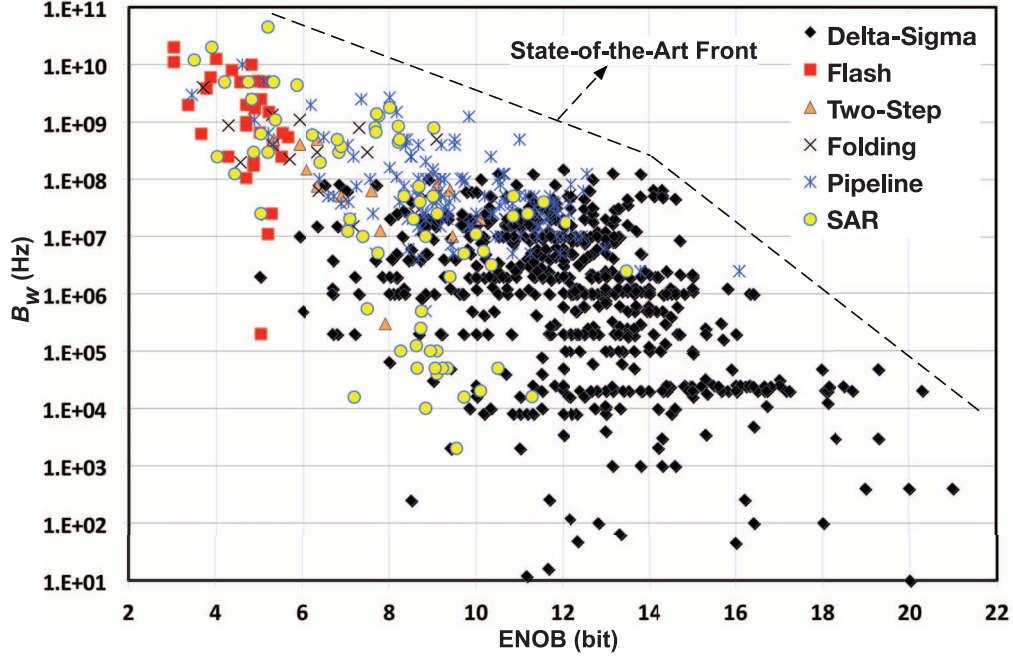


Fig. 1. Aperture plot of state-of-the-art ADCs.

The authors' views on the future of $\Delta\Sigma$ ADCs are given in Section V.

II. OVERVIEW OF THE STATE OF THE ART

In order to put $\Delta\Sigma$ Ms in the context of the state of the art on data converters, it is important to compare their performance with Nyquist-Rate ADCs. Murmann's database [11], which includes cutting-edge performance reported by ADCs presented at IEEE ISSCC and VLSI conferences, is complemented by the database compiled by de la Rosa [12], which considers also data reported in other major forums, such as ESSCIRC, CICC, A-SSCC and RFIC conferences. Both databases are updated up to March 2015, thus representing the starting point for this overview analysis.

A. Nyquist-Rate Versus $\Delta\Sigma$ ADCs

Fig. 1 represents the performance of cutting-edge ADCs in terms of their main design specifications, i.e., the digitized signal bandwidth, B_w , versus the resolution, characterized by the Effective Number Of Bits (ENOB). This plot—often referred to as *aperture plot*—illustrates the *conversion region* covered by the different kinds of ADCs. Note that a wide conversion region is covered by $\Delta\Sigma$ ADCs, comparable with that achieved by SAR converters. As depicted in the figure, two different state-of-the-art B_w -vs-ENOB fronts can be distinguished. One front is dominated by $\Delta\Sigma$ ADCs and goes from low-frequency (tens/hundreds of Hz) and high-resolution (<21 bit), to medium-high frequency (tens/hundreds of MHz) and medium-high resolution (<14 bit). The other front is dominated by SAR and Pipeline ADCs, and goes from hundreds of MHz and medium-resolution (in the order of 10 bit), to dozens of GHz and low-resolution (up to 5–6 bit). The ultimate limiting factors of both state-of-the-art fronts are thermal noise—in the

case of low-frequency, high-resolution specifications—and clock jitter—in those applications demanding the highest speeds.

It is clear from Fig. 1 that the state of the art is dominated by $\Delta\Sigma$, Pipeline and SAR ADCs. Apart from their main specifications, ENOB and B_w , it is instructive to compare the efficiency of these ADCs. This efficiency can be quantified by measuring the amount of *energy* per converted sample—also referred to as *conversion energy*, E , defined as [13]:

$$E \equiv \frac{P(W)}{f_{\text{snyc}}(\text{Hz})} \quad (1)$$

where P stands for the power dissipated and $f_{\text{snyc}} \equiv 2 \cdot B_w$, is the effective Nyquist rate.

Fig. 2 plots E as a function of the ENOB achieved by state-of-the-art $\Delta\Sigma$, Pipeline and SAR ADCs. This picture—commonly known as an *energy plot*—represents graphically the trade-off between resolution and conversion energy. This trade-off is usually quantified by a Figure of Merit (FOM), which takes into account the main performance metrics of an ADC, i.e., ENOB, B_w and P . The following two FOMs will be considered here:

$$\begin{aligned} \text{FOMW} &\equiv \frac{E(\text{J})}{2^{\text{ENOB}(\text{bit})}} \\ \text{FOMS} &\equiv \text{SNDR}(\text{dB}) + 10 \cdot \log_{10}[B_w(\text{Hz})/P(\text{W})] \end{aligned} \quad (2)$$

where SNDR stands for the Signal-to-(Noise and Distortion) Ratio and is related to ENOB as $\text{SNDR}(\text{dB}) = 10 \cdot \log_{10}[(3/2) \cdot 4^{\text{ENOB}(\text{bit})}] = 6.02 \cdot \text{ENOB}(\text{bit}) + 1.76$. FOMW, measured in J/conversion-step, was proposed by Walden [14], whereas FOMS is based on the FOM originally proposed by Rabii and Wooley [15], computed on a dB scale, as suggested by Schreier and Temes in [5], p. 357. FOMS can be written as:

$$\text{FOMS} \equiv P_{\text{sig|dBFS}} - \text{NSD|dBFS/Hz} - 10 \cdot \log_{10}[P(\text{W})] \quad (3)$$

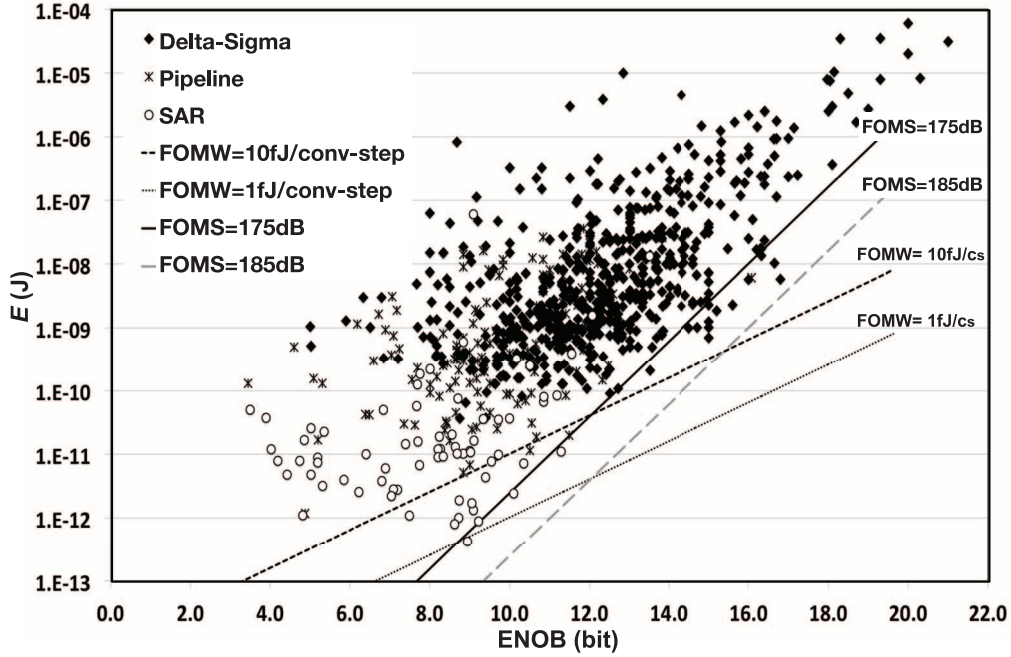


Fig. 2. Energy plot of state-of-the-art ADCs.

where $P_{\text{sig|dBFS}}$ is the input signal power computed in dB referred to the Full-Scale (FS) range of the converter, and $\text{NSD|dBFS/Hz} \equiv (P_{\text{nd}}/B_w)|_{\text{dBFS/Hz}}$ stands for the noise spectral density with P_{nd} being the noise plus distortion power referred to FS. Therefore, FOMS essentially characterizes the efficiency of a converter according to its NSD extrapolated to a 1-W power consumption when an FS input ($P_{\text{sig|dBFS}} = 0$) is considered. For example a converter with $\text{FOMS} = 160$ dB would have a $\text{NSD} = -160$ dBFS/Hz at $P = 1$ W, or $\text{NSD} = -150$ dBFS/Hz at $P = 0.1$ W.

As a reference, FOMW and FOMS are also depicted in Fig. 2, considering the following numerical values: $\text{FOMW} = 1$ and 10 fJ/conv-step; $\text{FOMS} = 175$ dB and 185 dB. The resulted lines of constant FOMW and FOMS in Fig. 2 can be respectively determined from (2) as:

$$\begin{aligned} E(\text{J})|_{\text{FOMW}} &= 2^{\text{ENOB}(\text{bit})} \cdot \text{FOMW} \\ E(\text{J})|_{\text{FOMS}} &= \frac{3}{2} \cdot \frac{4^{\text{ENOB}(\text{bit})}}{10^{\frac{\text{FOMS}}{10}}} \end{aligned} \quad (4)$$

Based on these metrics, it can be shown from Fig. 2 that SAR and $\Delta\Sigma$ ADCs are the most efficient ADC techniques, each one dominating different E -vs- ENOB regions. In terms of FOM, there are a number of SAR ADCs achieving a $\text{FOMW} < 10$ fJ/conv-step, and a few of them feature a FOMS slightly higher than 175 dB, while this value is exceeded by a number of $\Delta\Sigma$ ADCs with $\text{ENOB} > 14$ bit. Note also that the FOMS limit is more representative of the edge of the feasible region than FOMW. For this reason, FOMS should be preferred over FOMW in most cases.

B. Comparison of Different $\Delta\Sigma$ Techniques

In addition to comparing $\Delta\Sigma$ Ms with other kinds of ADCs, we can also compare various categories of $\Delta\Sigma$ ADCs, such as

single-loop (single-quantizer) or cascade topologies; Low-Pass (LP) or Band-Pass (BP); single-bit, multi-bit or Time-Encoding Quantization (TEQ); Continuous-Time (CT), either Gm-C or active-RC, or Discrete-Time (DT), mostly using Switched-Capacitor (SC) circuit techniques. Apart from these categories, an additional kind of $\Delta\Sigma$ Ms, referred to as hybrid $\Delta\Sigma$ Ms, has been also considered to include those topologies that combine diverse techniques, namely: CT and DT, active and passive circuits; $\Delta\Sigma$ /Nyquist-Rate ADCs, etc.

Fig. 3 represents the aperture plot of the state-of-the-art $\Delta\Sigma$ ADCs. Similarly to Fig. 1, two fronts can be identified. The first one covers a conversion region approximately limited by $1 \text{ MHz} < B_w < 100 \text{ MHz}$ and $10 \text{ bit} < \text{ENOB} < 15 \text{ bit}$, and it is dominated by CT- $\Delta\Sigma$ Ms, considering different architectures at GHz-range sampling rates—where clock jitter is one of the physical limits—including single-loop [16], cascade [17], [18] and BP/RF digitization [19]. The other front—mostly limited by thermal noise—covers high-resolution ($\text{ENOB} > 16 \text{ bit}$) with low-bandwidth ($B_w < 100 \text{ kHz}$) applications, where SC- $\Delta\Sigma$ Ms present the best metrics [20], [21].

Fig. 4 depicts the energy plot of state-of-the-art $\Delta\Sigma$ ADCs. Note that the most efficient designs are mainly based on CT circuits, some of them featuring a FOMS over 175 dB [22]–[26]. However, as will be discussed later on, the use of inverter-based circuit techniques has been successfully demonstrated in some single-bit SC- $\Delta\Sigma$ Ms, including both single-loop [8] and cascade [9] topologies. Moreover, the combined use of CT and SC techniques in the so-called hybrid- $\Delta\Sigma$ Ms, can lead also to energy-efficient solutions, as the design reported in [27], featuring $\text{FOMS} > 175$ dB in Fig. 4.

The $\Delta\Sigma$ ICs corresponding to the best data points in Figs. 3 and 4, define the *frontiers* of $\Delta\Sigma$ ADCs, showing the trends and challenges faced in diverse applications. Among

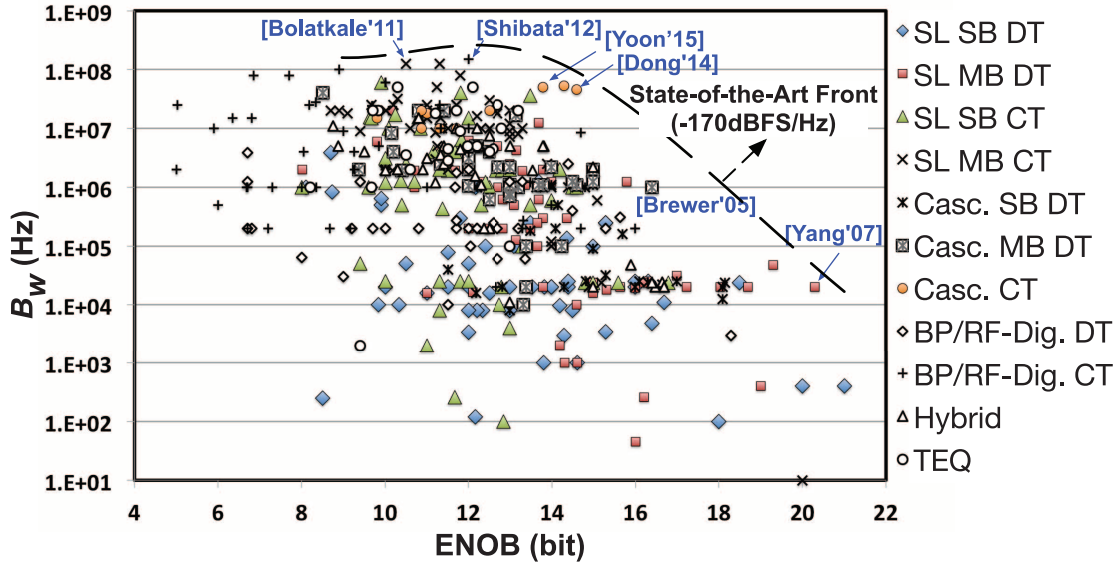


Fig. 3. Aperture plot of state-of-the-art $\Delta\Sigma$ Ms (SL = Single-Loop, SB = Single-Bit, MB = Multi-Bit, RF-Dig. = RF-to-Digital).

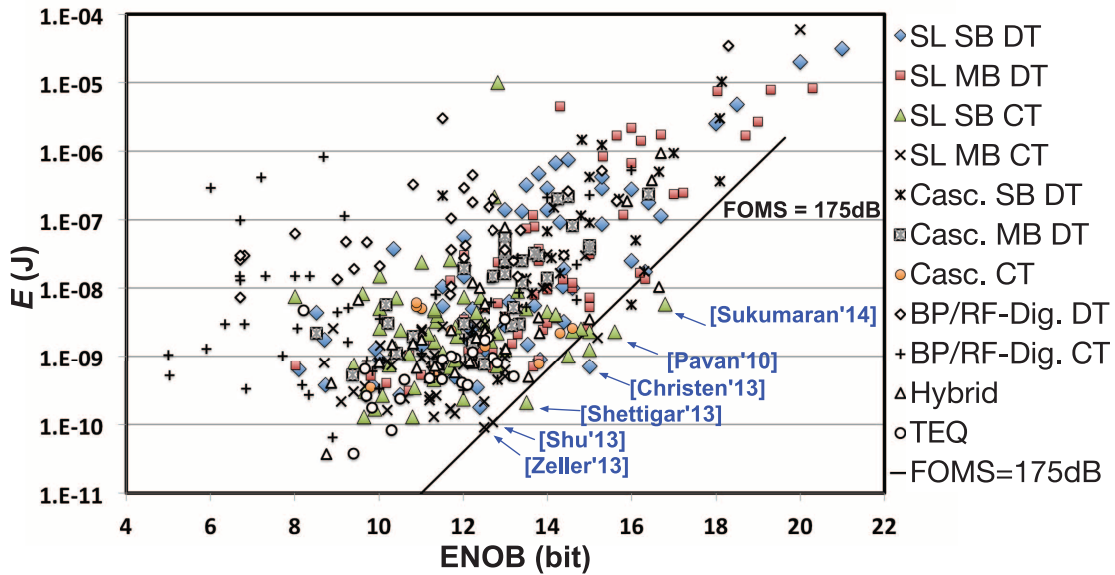


Fig. 4. Energy plot of state-of-the-art $\Delta\Sigma$ Ms.

others, the following circuits and systems techniques are included: digital-assisted analog circuits [8], [9], [25]; power-efficient analog filters and feedback Digital-to-Analog Converter (DAC) paths [23], [26]; mostly-digital/scaling-friendly quantization [24], [18]; RF/GHz-range [16], [19], etc. Some of these techniques are surveyed in the next sections. For the sake of clarity, the analyzed techniques are grouped according to the $\Delta\Sigma$ M building block in which they are mainly applied, either the feedforward loop filter, the quantizer and the feedback (DAC) path.

III. TRENDS IN THE DESIGN OF THE $\Delta\Sigma$ LOOP FILTER

The loop filter of $\Delta\Sigma$ Ms has evolved over the last few years by including a number of innovative techniques at both system- and circuit-levels, giving rise to new topologies and implementations. This section outlines some of these trends.

A. GHz-Range and RF-to-Digital $\Delta\Sigma$ Ms

Prompted by the higher transit frequencies in nanometer CMOS processes, CT- $\Delta\Sigma$ Ms are proving to be a competitive solution for the implementation of power-efficient ADCs operating in the GHz range [16], [19], [23], [28]–[35], thus opening the door to use them in *digital-intensive* RF transceivers [36] and SDR, sometimes referred to as $\Delta\Sigma$ receivers [37]. The most common approach in commercial mobile handsets is based on a direct conversion receiver like the one conceptually depicted in Fig. 5(a), which is made up of analog signal conditioning (basically an LNA, a mixer, and a baseband filter) followed by a LP CT- $\Delta\Sigma$ M. In addition to digitising the incoming signals, CT- $\Delta\Sigma$ Ms take advantage of their CT circuit nature to merge some RF functionalities like out-of-band blocker/interfering-rejection filtering, frequency-mixing process, channel-selection and antialiasing filtering [38]–[46]. All these functionalities

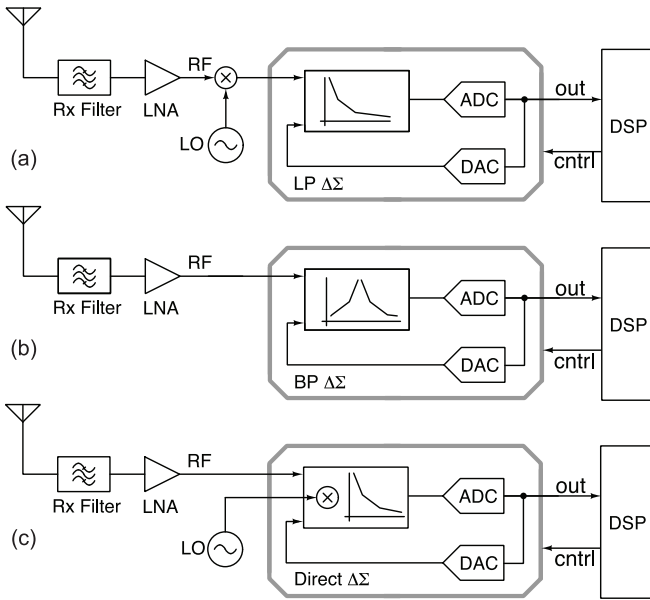


Fig. 5. Conceptual diagrams of different CT- $\Delta\Sigma$ RF receivers: (a) Direct conversion receiver based on a LP CT- $\Delta\Sigma$ with embedded Interfering-Rejection (IR) filter. (b) RF-to-digital converter based on a BP CT- $\Delta\Sigma$. (c) Direct $\Delta\Sigma$ receiver with embedded filtering and downconversion mixers.

can be embedded within the $\Delta\Sigma$ feedback loop as illustrated in Fig. 5(a), thus resulting in more compact RF receivers.

An alternative trend towards the practical implementation of SDR receivers is based on placing GHz-range BP CT- $\Delta\Sigma$ s as close as possible to the antenna, as illustrated in Fig. 5(b), yielding the so-called *RF-to-digital converters* [19], [32], [37], [47]–[54]. The majority of these $\Delta\Sigma$ -based RF *digitizers* include diverse strategies to reduce their aggressive specifications in terms of the sampling frequency (f_s) and the Dynamic Range (DR). Among others, the main strategies proposed include embedded out-of-band filtering [37], [53], frequency-translating [49], [53] and *undersampling* or *subsampling* [32], [47], [48]. The latter allows to digitize RF signals placed at $f_{RF} > f_s/2$, while keeping high values of the OverSampling Ratio (OSR), since the signal bandwidth is typically $B_w \ll (f_{RF}, f_s)$ [55], [56].

One of the first successful approaches using subsampling techniques was proposed by Ryckaert *et al.* [47], who reported a sixth-order *LC*-based BP- $\Delta\Sigma$ —depicted in Fig. 6(a)—for digitizing signals centered at 2.4 GHz with $f_s = 3$ GHz; i.e., a ratio of 4/5. To the best of the authors' knowledge, Martens *et al.* reported the highest sampling rate (8.8 GHz) to date by CMOS BP CT- $\Delta\Sigma$ s, digitising 2.2-GHz signals with 48-dB DR within 80-MHz, while consuming 164 mW. Such a sampling rate is achieved by combining six time-interleaved quantizers with a polyphase decimation filter [31].

The majority of reported RF-to-digital BP CT- $\Delta\Sigma$ s use a fixed center frequency or *notch frequency*. This forces the RF receiver to use a programmable frequency synthesizer in order to place the signal band within the pass-band of the BP- $\Delta\Sigma$. This issue has motivated the interest for reconfigurable/programmable BP- $\Delta\Sigma$ s with wide tunable notch frequency [57]. Thus, Shibata *et al.* reported a sixth-order reconfigurable

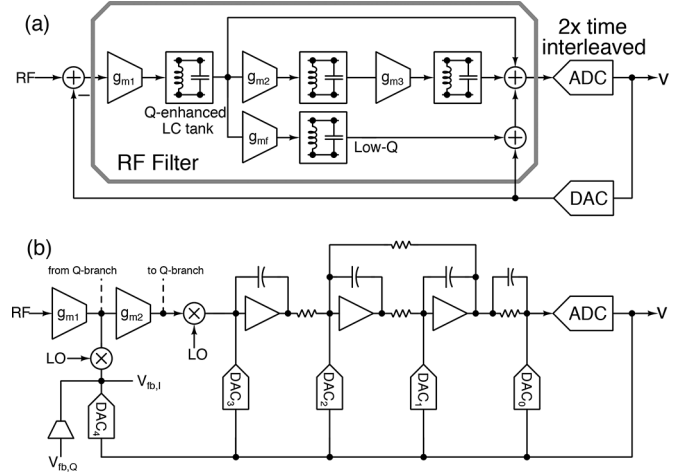
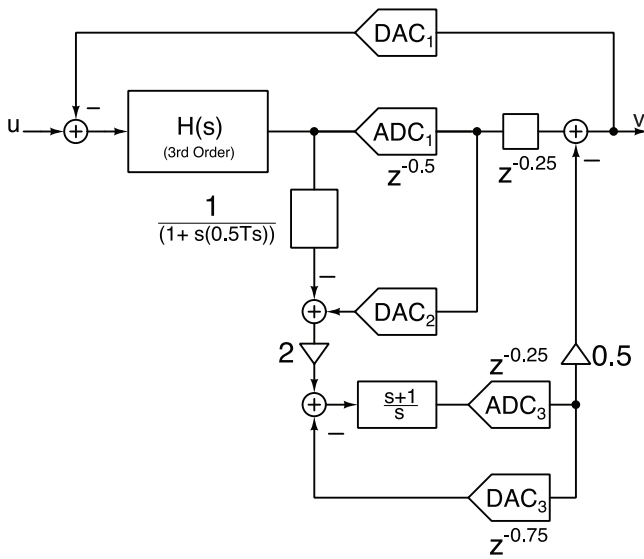


Fig. 6. Exemplary conceptual schematics of RF-to-digital CT- $\Delta\Sigma$ s. (a) Gm-LC BP- $\Delta\Sigma$ M reported in [47]. (b) Inductorless programmable $\Delta\Sigma$ receiver (1-branch) proposed in [7].

LP/BP- $\Delta\Sigma$ M which features a DC-to-1 GHz tunable notch frequency, while consuming 550 mW [19]. Another interesting approach based on a reconfigurable 0.8–2 GHz notch frequency was reported by Gupta *et al.* in [58]. In this case, a second-order CT BP- $\Delta\Sigma$ is integrated together with a Quadrature Phase-Locked Loop (QPLL) to allow quadrature phase synchronization between a raised-cosine feedback DAC and the embedded quantizer. The chip digitizes 1-MHz signals placed at 797 MHz with a DR of 8.3 bit, while consuming 41mW, out of which 11 mW are consumed by the QPLL [32].

The main challenge associated with the loop filters in BP- $\Delta\Sigma$ ADCs is achieving a high-quality and accurate resonance. For BP- $\Delta\Sigma$ ADCs intended for RF conversion, tuning range is also a concern. *LC* tanks are great from a power and linearity perspective but typically support only an octave of range, whereas active-*RC* resonators can be widely tunable but require amplifiers with high gain at the ADC's center frequency. These limitations have motivated exploring alternative signal processing techniques, such as translational circuits or N-path filters [59]. Although the idea of using N-path filters in BP $\Delta\Sigma$ s is not new [60], there has been a *resurgence* of these techniques to implement RF $\Delta\Sigma$ receivers. These techniques can be combined with either active or passive downconversion structures embedded into the modulator loop filter as conceptually shown in Fig. 5(c), in order to achieve the required received linearity and sensitivity [7], [37], [61]. Based on this idea, Wu *et al.* reported a 400 MHz-to-4 GHz direct RF-to-digital $\Delta\Sigma$ receiver implemented in 65nm CMOS, featuring SNDR > 68 dB within 4-MHz bandwidth across the carrier frequency band, while dissipating 40 mW [61]. More recently, Englund *et al.* presented an inductorless programmable 0.2–2.7 GHz $\Delta\Sigma$ receiver in 40 nm CMOS—conceptually depicted in Fig. 6(b), which is able to digitise 20-MHz signal bandwidths with SNDR = 40 dB and 90 mW power consumption [7]. Although the performance of these converters is still well short of what is needed for commercialization, this trend is set to continue and it is expected that $\Delta\Sigma$ receivers will be one of the key blocks for practical implementations of SDRs and next-generation mobile systems.

Fig. 7. SMASH 3-1 CT- $\Delta\Sigma$ M proposed in [18].

B. New Generations of Cascade $\Delta\Sigma$ Ms

In spite of the higher sampling rates demonstrated by some CT- $\Delta\Sigma$ Ms discussed above, new wireless standards, such as LTE, requires digitizing signals with $B_w > 100$ MHz, with no relaxation in noise density or power consumption [17]. This forces using low values of the OSR, thus requiring an increase in the number of bits of the embedded quantizer, B , and/or the order of the loop filter, L . The former strategy is limited by the power consumption of the embedded flash ADC, while the latter yields diminishing returns at low OSR. A well-known alternative consists of using cascade architectures. However, they can possess a high sensitivity to mismatch, which accounts for why the majority (>80%) of state-of-the-art $\Delta\Sigma$ Ms are single-loop topologies [6].

The mentioned limitations have motivated designers to look for alternative implementations of cascade $\Delta\Sigma$ Ms that reduce the sensitivity to noise leakage, thus benefiting from their aggressive noise-shaping capabilities. This is the case of the so-called *Sturdy* MASH (SMASH) $\Delta\Sigma$ M, proposed by Maghari *et al.* [62], in which the modulator output can be obtained from a direct digital addition (or subtraction) operation of the different stage outputs in the cascade, with no need for a Digital Cancellation Logic (DCL) as in conventional MASH topologies. The performance of SMASH DT- $\Delta\Sigma$ Ms can be notably improved if Unity STF (USTF) is used in all stages of the cascade [63], but the gain error between the inter-stage paths computing the front-end stage quantization error, e_1 , may degrade the modulator performance.

Although SMASH DT- $\Delta\Sigma$ Ms have been successfully implemented [64], the use of these architectures in wideband applications requires using CT circuits. However, their practical implementation requires a correct extraction of e_1 , which means that the input and the output of the front-end stage quantizer must be delayed by the same amount—something which is difficult to control in CT circuits.

Recently, Yoon *et al.* [18] reported a successful implementation of a SMASH 3-1 CT- $\Delta\Sigma$ M—conceptually depicted in

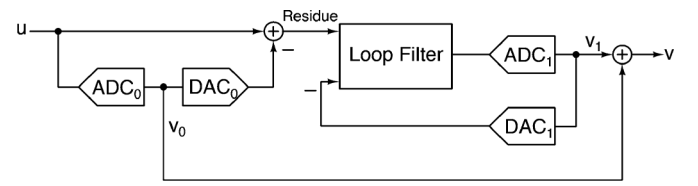
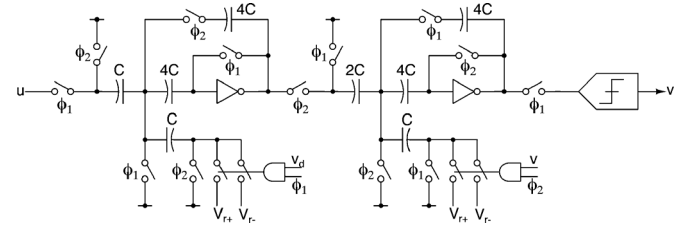
Fig. 8. Conceptual block diagram of a 0- L MASH $\Delta\Sigma$ ADC [65].Fig. 9. Inverter-based second-order single-loop $\Delta\Sigma$ M proposed in [66].

Fig. 7, which circumvents the mentioned limitations by using a LP filter in the inter-stage path and a feedforward path around the integrator of the second stage in order to obtain a USTF. The chip, fabricated in 28-nm CMOS, is clocked at 1.8 GHz and achieves SNDR = 74.6 dB within a 50-MHz signal bandwidth, while consuming 78 mW. These metrics place this SMASH CT- $\Delta\Sigma$ M in the frontier of the state of the art, according to the aperture plot shown in Fig. 3.

Another trend in the design of cascade $\Delta\Sigma$ Ms is based on a kind of two-step ADCs made up of a front-end flash ADC followed by a conventional $\Delta\Sigma$ M back-end stage, as conceptually depicted in Fig. 8. The resulting architecture is called 0- L MASH, since the front-end stage can be seen as a zeroth-order modulator—made up of a flash ADC, connected in cascaded by a back-end L -th order loop which digitizes the residue of the front-end stage [65]. The output of the ADC is given by the sum of the outputs of both sub-converters, $v = v_0 + v_1 = u + \text{NTF} \cdot (e_0 + e_1)$, where e_0 and e_1 are respectively the quantization errors of the front-end and back-end flash ADCs. Similarly to SMASH $\Delta\Sigma$ Ms, this topology does not need any DCL apart from a digital adder. Based on this strategy, Dong *et al.* [17] presented a 0-3 CT- $\Delta\Sigma$ M, which consists of the cascade topology of a 16-level flash ADC followed by a third-order active-RC $\Delta\Sigma$ M with an embedded 6-level flash ADC. The chip, integrated in 28-nm CMOS, does not need any additional DCL, obtaining an excellent performance as seen in Fig. 3, featuring SNDR = 71.4 dB within a 53.3-MHz signal band, and consuming 235 mW.

C. Inverter-Based $\Delta\Sigma$ Ms

Apart from the mentioned circuits and systems techniques, another efficient and elegant way of implementing the loop filter of $\Delta\Sigma$ Ms consists of replacing their most power-hungry building blocks, i.e., the Operational Transconductance Amplifiers (OTAs) by inverters, which are indeed one of the most power- and area-efficient OTA topologies. This circuit technique was first applied by Chae and Han in [66], who proposed an *inverter-based* second-order single-bit SC- $\Delta\Sigma$ M, whose conceptual schematic is shown in Fig. 9. This chip achieved 63-dB peak-SNDR within 8-kHz bandwidth, while consuming

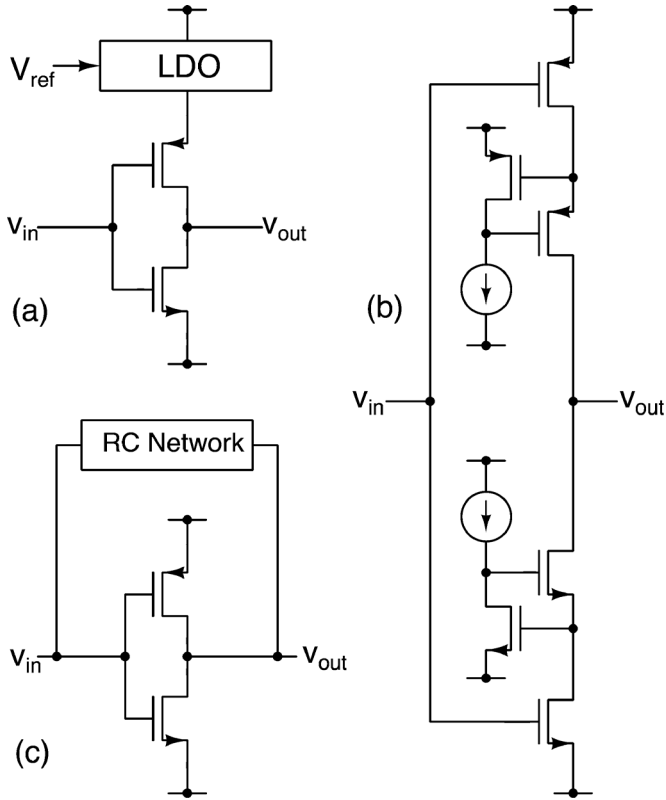


Fig. 10. Inverter-based OTAs used in state-of-the-art $\Delta\Sigma$ Ms. (a) Inverter with LDO-based biasing [8]. (b) Gain-boosted class-C inverter [9]. (c) Third-order inverter-based integrator used in the CT- $\Delta\Sigma$ M reported in [25].

only $5.6 \mu\text{W}$ from a 1.2-V supply (FOMS = 156 dB). Although the PSRR was not good, these results demonstrated that the use of *digital-inspired* analog circuit techniques may become very energy efficient while keeping an outstanding performance.

In spite of the potential benefits of replacing conventional OTAs by CMOS inverters, it should be noted that these simple circuits cannot be seen as the *panacea* for all analog design problems in $\Delta\Sigma$ Ms. Among others, CMOS inverters achieve a higher power efficiency at the price of sacrificing other performance metrics such as robustness over PVT variations, impact of circuit parasitics, limited dc gain, etc. These limitations have motivated some authors to proposed improved versions of inverted-based integrators to implement the loop filter of $\Delta\Sigma$ Ms [8], [9], [25], [67].

Christen proposes an inverter-based OTA where the operating point is controlled over the internal supply voltage by using a Low-DropOut (LDO) regulator, as conceptually depicted in Fig. 10(a) [8]. This technique is applied to the design of a fourth-order feedforward single-loop single-bit SC- $\Delta\Sigma$ M which digitizes 20-kHz signals with SNDR = 87.9 dB, while consuming $140 \mu\text{W}$. These metrics yield the lowest value of E reported so far by SC- $\Delta\Sigma$ Ms, as illustrated in Fig. 4. Another cutting-edge inverter-based SC- $\Delta\Sigma$ M is reported by Luo *et al.* [9], who propose a gain-boosted class-C inverter—depicted in Fig. 10(b). This technique allows to increase the dc gain of the inverter-based OTA over 80 dB, being used to implement a cascade 2-1 SC- $\Delta\Sigma$ M, featuring 91-dB SNDR within a 20-kHz signal bandwidth with $230 \mu\text{W}$ power dissipation.

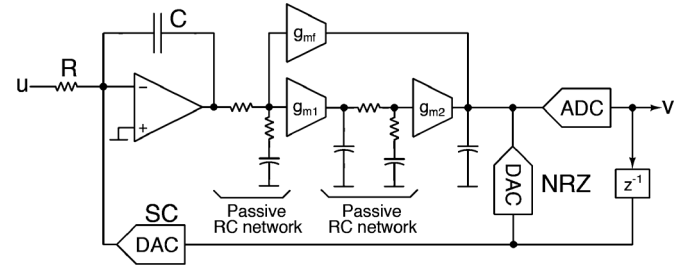


Fig. 11. Example of the hybrid active-passive CT- $\Delta\Sigma$ M reported in [68].

Inverter-based OTAs have been also demonstrated in CT- $\Delta\Sigma$ Ms, as shown by Zeller *et al.* [25], who used an inverter-based single-opamp third-order filter—conceptually depicted in Fig. 10(c)—in a pseudo-differential single-loop 1.5-bit CT- $\Delta\Sigma$ M featuring 68.6-dB SNDR over a signal bandwidth of 10 MHz while consuming 1.82 mW. Thus, as demonstrated by these outstanding designs, the use of inverter-based loop filters are very promising strategies for the implementation of high-performance $\Delta\Sigma$ Ms, and it is expected that the use of these circuit techniques might benefit from technology downscaling in the next few years.

D. Hybrid Active/Passive and Amplifier-Less $\Delta\Sigma$ Ms

Another alternative to reduce the number of OTAs used in $\Delta\Sigma$ Ms consists of replacing some active building blocks with passive-RC networks $\Delta\Sigma$ Ms [68]. In addition to reducing the power consumption, integrated passive networks feature potentially better linearity than their active counterparts, at the price of reducing the required gain to make the loop filter robust enough against thermal noise and other circuit nonidealities. These limitations can be partially mitigated by using a hybrid active/passive loop-filter, in which the front-end subcircuit (either an integrator or a resonator) is active, so that the effect of thermal noise of the passive circuit elements can be reduced by the preceding active blocks. Prompted by the mentioned potential benefits, passive RC networks are becoming a popular circuit solution for the implementation of both CT- $\Delta\Sigma$ Ms and SC- $\Delta\Sigma$ Ms [30], [54], [68]–[72].

Fig. 11 shows one of the first successful examples of this idea applied to CT- $\Delta\Sigma$ Ms, which was proposed by Song *et al.* in [68]. The modulator—consisting of a fifth-order single-loop architecture—uses five integrators, two of them implemented by passive RC networks. These passive integrators are placed in the second and fourth position of the modulator chain, so that their nonideal effects can be attenuated by the gain of active integrators. Another interesting example was proposed by Srinivasan *et al.* [30], who reported a third-order CT- $\Delta\Sigma$ M made up of a front-end passive-RC integrator followed by two Gm-C integrators. The chip—fabricated in a 45-nm CMOS technology—digitizes 60-MHz signals with 61-dB SNDR and 20 mW power consumption, while clocked at 6 GHz.

Although hybrid active/passive loop filters have been mostly used in LP- $\Delta\Sigma$ Ms, these techniques can be also applied to BP- $\Delta\Sigma$ Ms. Due to their higher operating frequencies, these modulators present larger sensitivity to circuit element tolerances and technology process variations, as well as the

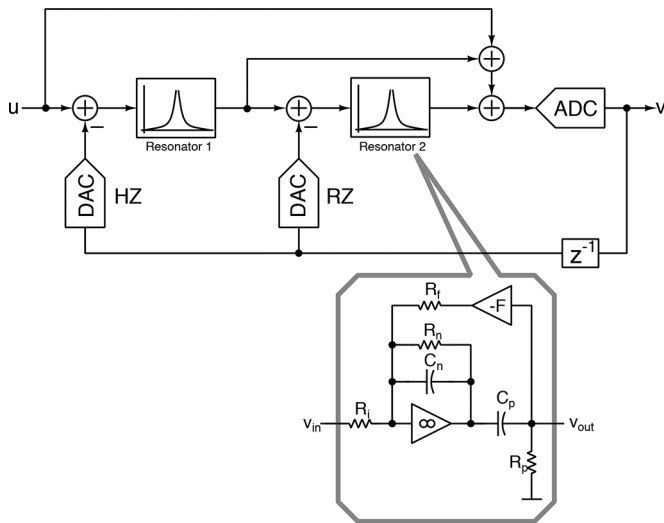


Fig. 12. BP CT- $\Delta\Sigma$ M proposed in [54] based on single-opamp resonators with Q -factor boosting.

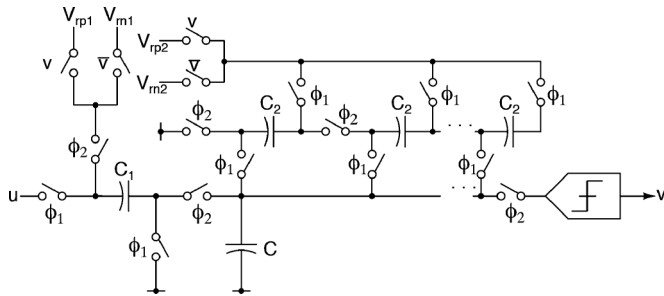


Fig. 13. Passive second-order SC- $\Delta\Sigma$ M proposed in [70] (single-ended version).

degradation caused by parasitics. For instance, Chae *et al.* [54] reported a fourth-order BP CT- $\Delta\Sigma$ M which digitizes 24-MHz signals located at 200 MHz with 58-dB SNDR while consuming 12 mW. The chip is implemented by using single-opamp resonators, which incorporate positive feedback to achieve higher quality factors, and are implemented by combining an active LP Filter (LPF) and a passive High-Pass Filter (HPF) as illustrated in Fig. 12.

Apart from being used in wideband CT- $\Delta\Sigma$ Ms, passive networks have been also applied to SC- $\Delta\Sigma$ Ms intended for ultra-low-power applications. A good example of the latter is the work reported by Yeknami *et al.* [70], who proposed several alternative (hybrid active/passive and fully-passive) implementations of a second-order single-bit SC- $\Delta\Sigma$ M intended for medical implant devices. As an illustration, Fig. 13 shows the conceptual schematic of a passive SC- $\Delta\Sigma$ M, which digitizes 500-Hz signals with 65-dB SNDR, while consuming just 0.43 μ W.

This trend is expected to continue, being a very promising alternative technique in application scenarios where energy efficiency is a must. Indeed, hybrid active/passive circuits could be properly combined with other kinds of hybrid circuit techniques—such as the so-called CT/SC $\Delta\Sigma$ Ms—to take advantage of the different circuits involved, i.e., faster operation of

CT circuits and lower sensitivity of SC circuits to errors [27], [73]–[78].

E. Power-Efficient Loop-Filter Opamp Techniques

It is well known that the bandwidth requirements of loop-filter opamps in CT- $\Delta\Sigma$ Ms are greatly reduced when compared with their SC- $\Delta\Sigma$ M counterparts, which is what makes CT loop-filter techniques generally more efficient as shown in the energy plot of Fig. 4. Indeed, the majority of CT- $\Delta\Sigma$ Ms achieving the best performance in that figure use a loop filter based on a Cascade of Integrators with Feed-Forward (CIFF) topology with single-bit quantization [22], [23], [26], [79].

However, apart from the obvious benefits in terms of the increased number of bits, multibit quantization is also appealing in practical broadband applications demanding low OSR, with the additional benefit of reducing the sensitivity to clock jitter, at the price of including linearization techniques in the feedback path to compensate for the DAC nonlinearities, with the subsequent penalty in power consumption and circuit complexity. A detailed study about the different $\Delta\Sigma$ loop-filter topologies is beyond the scope of this paper and can be found in a number of references [5], [6], [46], [79], [80]. Instead, some key circuit techniques used by most efficient CT- $\Delta\Sigma$ Ms to implement their most power-hungry building blocks, i.e., the opamps, are outlined.

Multi-stage Feed-Forward (FF) compensated opamps have been used in some state-of-the-art CT- $\Delta\Sigma$ Ms [81], because they are more power-efficient than other topologies—such as Miller-compensated structures—since they do not waste current in charging and discharging the (explicit) compensation capacitor [23]. The main problem associated with FF-compensated opamps is their long settling time, which is a limiting factor in SC- $\Delta\Sigma$ Ms. However, this is not important in CT- $\Delta\Sigma$ Ms, where the entire integrator output waveform is processed by the rest of the loop filter.

Another power-efficient technique is the so-called *assisted-opamp* integrator, proposed by Pavan *et al.* [22] and conceptually illustrated in the single-bit CT- $\Delta\Sigma$ M shown in Fig. 14. Essentially, the problem is that the opamp in the front-end integrator cannot respond fast enough to the sharp steps of the 1-bit DAC feedback voltage. This effect causes large jumps in the virtual ground potential of the front-end opamp whenever the DAC switches—regardless of the DAC waveform used. The main consequence of these voltage jumps is nonlinearity which severely degrades the performance of the modulator, unless large bias currents are used to improve the dynamics of the opamp. Instead, an assisted-opamp integrator includes an additional (*assistant*) transconductor and a replica of the feedback DAC to pull current out of the OTA, thus reducing the virtual ground voltage excursions.

Assisted opamps have been successfully used in the front-end integrator, which is usually the most power-consuming building block in a $\Delta\Sigma$ M when a single-bit quantizer, i.e., a comparator, is used. Indeed, this technique can be applied to different OTA topologies [22], [23], including two-stage feedforward compensated and Miller-compensated OTAs, featuring state-of-the-art

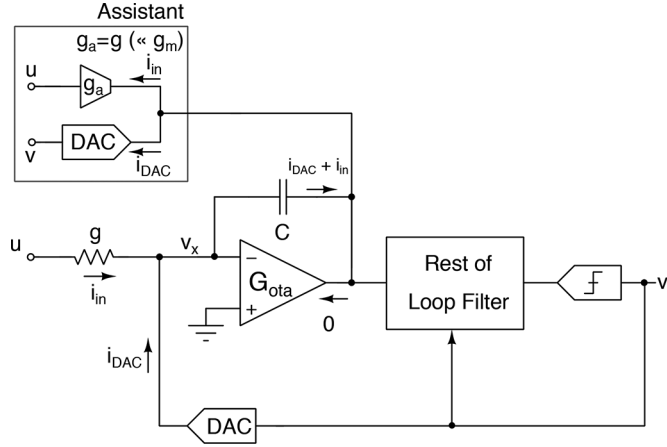


Fig. 14. CT- $\Delta\Sigma$ M with assisted-opamp integrator proposed in [22].

performance with SNDR = 88 dB within 24-kHz signal bandwidth with 110 μ W.

IV. TRENDS IN THE DESIGN OF THE $\Delta\Sigma$ QUANTIZER

As stated in previous sections, in addition to the loop-filter itself, there have been a number of innovative techniques in the design of the embedded quantizer and feedback DAC path, giving rise to a number of new kinds of $\Delta\Sigma$ Ms described below.

A. Hybrid $\Delta\Sigma$ M/Nyquist-Rate ADCs

Based on the Leslie-Singh architecture [82], Brooks *et al.* [83] reported one of the first attempts to combine the benefits of Nyquist-Rate and $\Delta\Sigma$ Ms in a single ADC, by cascading a 5-bit second-order $\Delta\Sigma$ M and a 12-bit fourth-stage pipeline ADC. Since then, there has been a number of hybrid $\Delta\Sigma$ M-Nyquist ADCs—considering both single-loop and cascade topologies—featuring a competitive performance in different application scenarios [84]–[94].

The so-called incremental $\Delta\Sigma$ ADCs, can be also considered as hybrid $\Delta\Sigma$ M/Nyquist-rate ADCs. These kinds of ADCs use oversampling and noise-shaping, but reset the loop-filter integrators and digital filters after each conversion. These features allow these converters to be especially efficient in low-frequency high-resolution applications like sensor interfaces [95] and other applications requiring to process multiplexed low-frequency signals with high-resolution [96].

In the majority of cases, the basic strategy followed by hybrid $\Delta\Sigma$ M/Nyquist ADCs consists of replacing the multibit flash ADC embedded in the $\Delta\Sigma$ M, by another kind of Nyquist-rate ADC architecture, namely: pipeline [83], [84], [88], [91], two-step flash [88], SAR [86], [90], [92]–[94], cyclic [85], [87] and integrating ADC [89]. The main advantage of this strategy is to provide a way of increasing the number of levels of the internal quantizer without the prohibitive exponential growth of the power consumption and silicon area of conventional flash ADCs.

One of the main inconveniences of the combination of $\Delta\Sigma$ Ms and Nyquist-rate ADCs is that, as the number of bits of the Nyquist-rate sub-ADC increases, the circuit complexity, sensitivity to circuit errors, and latency increases as well, with the risk of losing the benefits provided by $\Delta\Sigma$ Ms in terms of

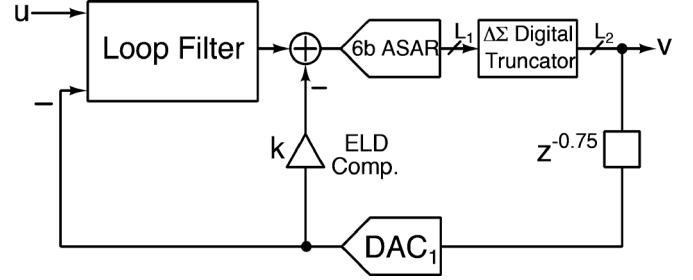


Fig. 15. Hybrid ASAR-based third-order CT- $\Delta\Sigma$ M proposed in [90].

system simplicity and robustness against circuit nonidealities. Therefore, even though very competitive performance can be achieved by replacing multibit flash ADCs with other alternative Nyquist-rate ADCs, the mentioned design trade-offs have motivated $\Delta\Sigma$ M designers to explore alternative implementations to conventional multibit quantization.

Among other approaches, the use of an Asynchronous SAR (ASAR) structure as the embedded quantizer ADC is gaining favor because their potential better efficiency [90], [93], [94]. Fig. 15 shows the conceptual diagram of the hybrid ASAR-based CT- $\Delta\Sigma$ M proposed by Tsai *et al.* [90]. The chip, integrated in 40-nm CMOS, consists of a third-order loop-filter with a 6-bit ASAR quantizer followed by a digital $\Delta\Sigma$ truncator, which reduces the ASAR output level from $L_1 = 64$ to $L_2 = 9$ to achieve an optimum performance, featuring SNDR = 79.6 dB within a 1.92-MHz signal bandwidth with 1.91 mW. A similar approach is followed by Wang *et al.* [93], by embedding a 6-bit ASAR together with digital truncation and digital integration in a second-order CT- $\Delta\Sigma$ M implemented in 28-nm CMOS, getting a FOMS = 173.8 dB. More recently, Ho *et al.* [94] reported a 4th-order CT- $\Delta\Sigma$ M with 4-bit ASAR with Data-Weight Averaging (DWA) to compensate for the nonlinearity of the multi-bit feedback DAC, obtaining SNDR = 90.4 dB, $B_w = 2.2$ MHz and 4.5 mW.

Thus, the combination of Nyquist-Rate ADCs—especially SAR ADCs—and $\Delta\Sigma$ Ms is giving rise to new kinds of ADC architectures, which are in general more *digital-friendly* than conventional ADCs. A good example of this is the SAR-based $\Delta\Sigma$ ADC reported by Sanyal *et al.* [92], conceptually depicted in Fig. 16. The ADC can be seen as a two-stage 0–1 MASH stated earlier, and uses a similar concept to the so-called *zoom*-ADC proposed by Soury and Makinwa in [97], [98]. In the ADC of Fig. 16, the first stage is a 5-bit SAR which performs the coarse quantization of the input signal and its residue is digitized by a first-order $\Delta\Sigma$ M mostly based on a Voltage-Controlled-Oscillator (VCO), without needing OTAs for its implementation. Indeed, the use of time/frequency-based quantization is another approach attempted by $\Delta\Sigma$ M designers in recent years as discussed below.

B. Time/Frequency-Based Quantization

Regardless of whether the $\Delta\Sigma$ M quantizer is implemented by a conventional flash ADC, a tracking structure [24], [99], or any other Nyquist-rate ADC stated above, the use of amplitude-based linearization becomes a design challenge due to the reduction in voltage supplies associated with CMOS technology

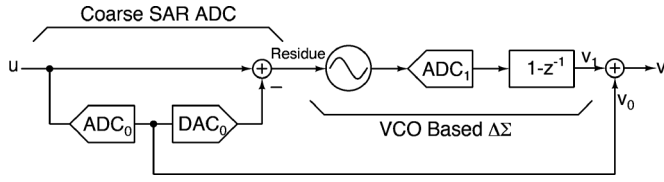


Fig. 16. SAR-VCO ΔΣ ADC proposed in [92].

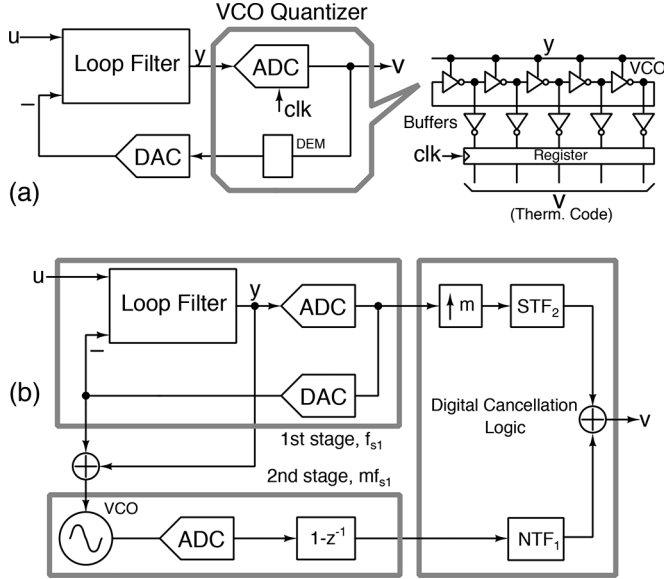


Fig. 17. Conceptual diagram of VCO-based ΔΣMs: (a) Single-loop topology proposed in [101]. (b) Multi-rate cascade presented in [112].

downscaling. This problem has motivated the exploration of alternative quantizer implementations, based on time/frequency coding instead of amplitude coding. The benefit of this approach is that the resulting quantization methods are more suited for the implementation in low voltage (<1 V) nanometer CMOS processes, although special care must be taken in terms of those errors which are directly related to the timing, such as clock jitter error in CT-ΔΣMs [100].

Among others, the use of VCO-based quantizers has been the most common approach to replace conventional (amplitude-based) multi-bit quantizers [36], [101]–[111]. The idea is based on translating the quantized information in the amplitude domain to the time domain by means of a voltage-to-frequency conversion, which can be performed by a VCO circuit as conceptually depicted in Fig. 17(a), which shows one of the first implementations of this idea in ΔΣ ADCs [101]. The principle of operation behind this approach relies on the use of a ring oscillator to count the number of edges within a given time period. The result is directly related to the input signal, thus obtaining a digital representation of the amplitude.

The main limitation of VCO-based quantization is the inherent nonlinearity of the voltage-to-frequency conversion, necessitating the use of either calibration or linearization techniques [107]–[109], [113]. Some authors propose also to mitigate this limitation by using a phase detector instead of a frequency detector as proposed in [102], which provides also a first-order noise shaping. Another strategy consists of embedding VCO-based quantizers in the back-end stage of a cascade

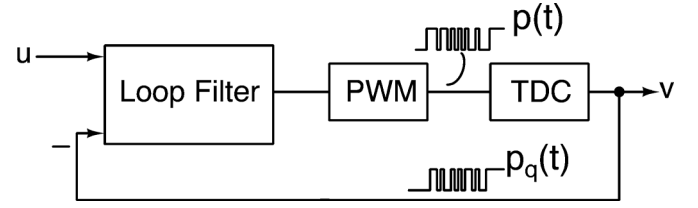


Fig. 18. PWM-based ΔΣM concept reported in [114].

ΔΣM, such that their nonlinearities can be attenuated by the noise shaping of precedent stages in the cascade. One of the first implementations of this idea, proposed by Zaliasl *et al.* [112], consists in a two-stage cascade multi-rate ΔΣM, illustrated in Fig. 17(b), in which the back-end stage (operating at a higher rate) is implemented by a VCO quantizer, thus benefiting from its potentially higher operation speed without being penalized by its intrinsic nonlinearity. Indeed, this strategy is also behind the two-step hybrid Nyquist-rate/ΔΣ ADC proposed in [92] and discussed in previous section.

An alternative to VCO-based quantizers consists of using pulse-width modulation (PWM) [114], [115], as conceptually depicted in Fig. 18. The PWM block converts the voltage at the output of the loop filter to a pulse $p(t)$ with corresponding width and a time-to-digital converter (TDC) generates digital codes that are discrete representations of the edges of the $p(t)$ signal; i.e., a time-quantized representation. Indeed, embedded TDCs have been successfully implemented by a number of recent ΔΣMs [116]–[119].

One of the main problems of TDCs and PWM-based quantizers is their limited dynamic range [100]. This issue can be addressed by using the so-called *time-encoding quantizer* (TEQ) proposed by Hernandez *et al.* [120]. Essentially, the idea underlying TEQ-based ΔΣMs consists of using a modulator whose loop filter is split into two loops, one of which is made to oscillate, thus providing an independent control of the oscillation as a difference with PWM-based quantization schemes. Based on this idea, Prefasi *et al.* [121] reported a third-order CT-ΔΣM integrated in 65 nm CMOS and clocked at 2.56 GHz, featuring a SNDR = 61 dB within 20-MHz bandwidth.

C. Mostly-Digital ΔΣMs

Time/frequency-based circuit techniques have not only been used for replacing quantizers, but also for other main building blocks of ΔΣMs. One of the first attempts to implement this idea was proposed by Taylor and Galton in [122], who used a Voltage-Controlled Ring Oscillator (VCRO) as the essential block of the ΔΣ ADC. The proposed modulator—mostly implemented by digital circuit techniques—consists of a digitally background calibrated VCRO, in which its inverters are sampled at the required output sample-rate, without containing any analog integrators, comparators, or feedback DACs. The main advantage of this approach is that, as the ΔΣM is essentially implemented by digital circuits, it may benefit from the CMOS technology downscaling in terms of speed, with the subsequent benefits for low-voltage low-power wideband applications. Moreover, the operation of VCRO-based ΔΣMs essentially depends on the speed of its digital blocks, thus achieving high

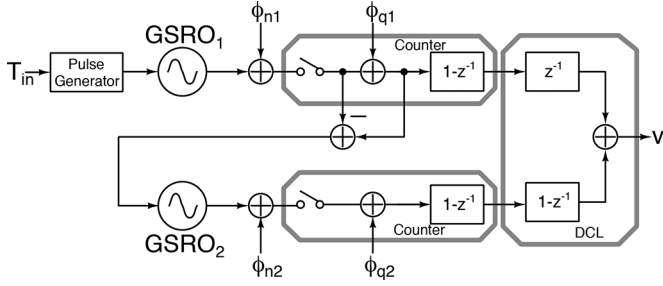


Fig. 19. Conceptual model of the mostly-digital GSRO-based 1-1 MASH $\Delta\Sigma$ TDC proposed in [10].

accuracy via digital signal processing. However, the main limitation of this approach is its higher sensitivity to timing errors, such as clock jitter error, and the aforementioned nonlinear behavior of VCOs. These problems are overcome in [123] by using digital background calibration and self-canceling dither techniques, yielding to state-of-the-art performance results, $\text{SNDR} = 70 - 75$ dB within $B_w = 5 - 37.5$ MHz with a tunable sampling rate of 1.3–2.4 GHz and a FOMS = 160 dB [124].

More recently, Yu *et al.* [10] propose a 1-1 MASH $\Delta\Sigma$ TDC—conceptually depicted in Fig. 19—based on the use of the so-called Gated Switched-Ring Oscillator (GSRO). The latter, which is essentially a voltage-controlled SRO, introduces the required synchronization between both stages of the modulator, so that no calibration is needed to compensate for the mismatch caused by the frequency differences between both oscillators in the cascade.

Following the same philosophy of using VCOs as basic building blocks, Lee *et al.* [125] presented an OTA-less CT- $\Delta\Sigma$ implemented with both VCO-based loop-filter and a quantizer, leading to a mostly-digital scaling-friendly ADC, which features a FOMS = 156 dB. Another innovative architecture, consisting of a third-order CT- $\Delta\Sigma$ with a loop filter based on VCO-based integrators was presented by Young *et al.* [126]. The chip operates at 1.28 GS/s, achieving an $\text{SNDR} = 64$ dB within 50-MHz bandwidth with 38mW power dissipation. Therefore, although the performance of these VCO-based $\Delta\Sigma$ s is mainly limited by nonlinearity, these designs demonstrate that their metrics can be improved in deeper nanometer CMOS—28 nm and beyond.

D. Finite Impulse Response (FIR) in the $\Delta\Sigma$ Feedback Path

As discussed earlier, the use of multi-bit quantization presents a number of inconveniences in terms of circuit complexity, area/power consumption and the inherent nonlinearity caused by mismatches in the feedback DAC elements. Indeed, since a $\Delta\Sigma$ ADC is a feedback system, the performance of the ADC can be no better than that of its internal feedback DAC. A number of linearization strategies have been proposed in the past to compensate for this error, such as Dynamic Element Matching (DEM), calibration, or the use of auxiliary DACs, which are based on a look-up-table computation of the INL, assists the main feedback DAC [127]. In the majority of cases, these techniques introduce additional circuitry complexity, thus

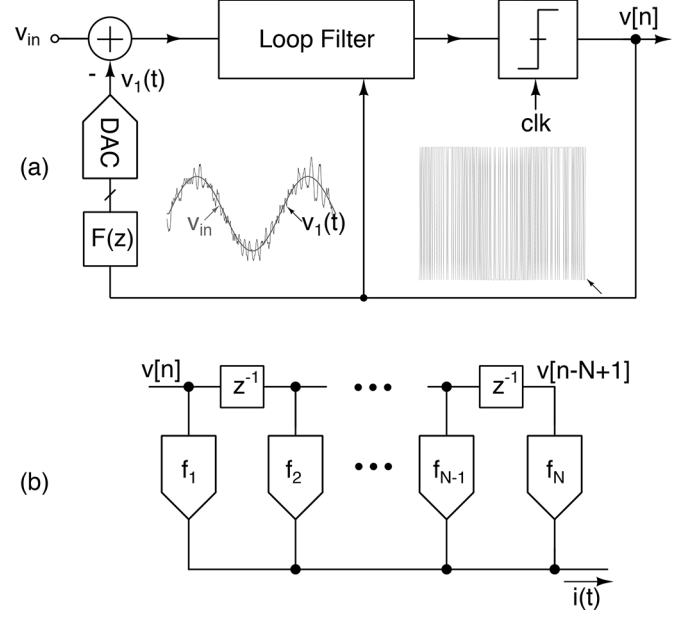


Fig. 20. FIR-DAC based single-bit CT- $\Delta\Sigma$ Ms: (a) $\Delta\Sigma$ M, (b) FIR filter.

limiting the maximum resolution of the embedded quantizer to typically less than 4–5 bit.

To address this problem, some authors propose the use of alternative implementations of the modulator feedback waveforms—such as a DAC with Finite Impulsive Response (FIR) [23], [26], [30], [128]. The idea—originally proposed by Oliaei [129] and conceptually illustrated in Fig. 20—is to feedback a filtered version of the single-bit quantization output, in such a way that due to the high-frequency attenuation of the FIR filter, the DAC output is a multi-level waveform. This way, using a single-bit ADC and a FIR DAC allows to obtain the low-jitter sensitivity and high linearity of a multi-bit $\Delta\Sigma$ M, while keeping the simplicity and robustness of single-bit $\Delta\Sigma$ Ms.

Note that FIR DACs introduce an extra delay in the modulator loop filter. This delay has to be properly controlled to stabilise the system and guarantee a correct performance. This problem can be solved by using the methods of moments, which allows us to perfectly compensate for the FIR DAC's delay as demonstrated in [130]. Another issue is related to the number of taps used in the FIR filter. This can be optimised to maximise the modulator performance and efficiency as shown by Sukumaran and Pavan [26], achieving one of the most efficient designs reported to date according to the energy plot shown in Fig. 4, featuring FOMS = 178 dB ($B_w = 24$ kHz, DR = 99 dB, $P = 0.3$ mW).

E. DAC Techniques for High-Performance CT- $\Delta\Sigma$ Ms

With the trend toward high-speed CT- $\Delta\Sigma$ Ms, nonidealities in the feedback DAC, especially dynamic errors, are becoming dominant. Although mismatch-shaping can be used to alleviate the errors associated with static element errors and static timing errors, mismatch-shaping increases the switching activity and therefore exacerbates the degradation caused by dynamic errors. Risbo *et al.* [131] have shown that it is possible to address this problem with an elaborate element selection logic which

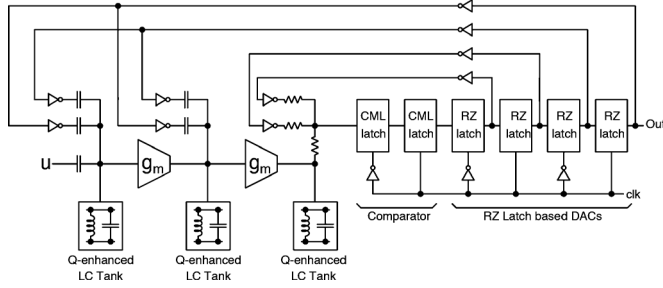


Fig. 21. BP CT- $\Delta\Sigma$ M based on Q -enhanced LC tanks and voltage-mode RZ latched DACs proposed in [29].

simultaneously shapes both mismatch-induced noise and the noise associated with the dynamic error commonly referred to as Inter-Symbol Interference (ISI). The scheme was quite practical for a clock rate of 3 MHz, but would be difficult to implement at the GHz clock rates typical of current high-speed CT- $\Delta\Sigma$ ADCs. Consequently, other approaches, such as ISI calibration via analog adjustment or digital correction, are likely to be used in future high-speed CT- $\Delta\Sigma$ ADCs.

Current-mode DACs are typically used inside high-speed CT- $\Delta\Sigma$ ADCs, but resistor DACs [132] are preferred when noise is the most critical specification. To see why, note that the current noise of a MOS device relative to its output current is inversely proportional to the overdrive voltage, $V_{gs} - V_T$, whereas the current noise of a resistor relative to its output current is inversely proportional to the voltage across the resistor, with proportionality factors of $(16/3)kT$ and $4kT$, respectively. Since $16/3 > 4$ and since the value of $V_{gs} - V_T$ is typically less than $V_{DD}/2$, a resistor switched between 0 and V_{DD} has less noise than an MOS current source. Sukumaran and Pavan [26] describes a $\Delta\Sigma$ ADC using resistor DACs that achieves an outstanding performance as detailed earlier. Single-bit quantization is used to minimize power and an FIR DAC is used to counter the jitter sensitivity normally associated with single-bit quantization. Further minimization of DAC noise is possible by using tri-level (0, I , $-I$) DAC elements, or by using voltages outside the supply rails.

In addition to DACs built with MOS current sources or switched resistors, CT- $\Delta\Sigma$ ADCs can also make use of switched-capacitor DACs. These DACs were originally used to reduce the jitter sensitivity of LP ADCs, but more recently have been used by Harrison *et al.* [29] as voltage feedback to the Q -enhanced LC tanks of a BP ADC, as conceptually illustrated in Fig. 21. In this case, inverters are used as voltage-mode DACs, driven by Return-to-Zero (RZ) latches. In the original usage scenario, the fact that the charge supplied by a switched capacitor is insensitive to the clock period yielded the desired jitter-insensitivity. However, since the high peak current associated with the switched capacitor tends to require a more powerful amplifier than if the DAC current is steady, this technique has fallen out of favor. The more recent usage effectively implements voltage feedback to the LC tank, which is useful since current feedback alone provides only one degree of control over a second-order system.

V. DISCUSSION AND CONCLUSION

$\Delta\Sigma$ converters encompass a rich variety of architectures and circuit techniques including single-/multi-bit, discrete-/continuous-time, single-/multi-loop, lowpass/bandpass/quadrature, etc. Over the last years, we have witnessed a steady refinement of $\Delta\Sigma$ topologies, fuelled by a rapid expansion of novel circuit strategies to realize the loop-filter functions and new dimensions in the signal quantization and reconstruction.

While yesterday's workhorse was the single-bit single-loop lowpass system built with switched-capacitor circuitry, the demand for more bandwidth and reduced power consumption, the pressures of nm-scale MOS devices and the development of mismatch-shaping and feedforward amplifier techniques have shifted the dominant implementation towards multi-bit $\Delta\Sigma$ modulators with continuous-time loop filters. Multi-stage architectures employing continuous-time loop filters have recently demonstrated further increases in bandwidth and are expected to be widely commercialized.

The overview of the state of the art on $\Delta\Sigma$ ADCs carried out in this work, has highlighted these and other emerging topics, design trends and circuits and systems approaches which are in the frontiers of data converters. Among others, RF-to-digital and GHz-range $\Delta\Sigma$ M; robust cascade architectures; digital-based, hybrid active/passive and power-efficient loop-filter circuit implementations; Nyquist-rate/ $\Delta\Sigma$ ADCs, time-based quantization and high-performance feedback DACs have been outlined, identifying outstanding designs in each case. These trends are expected to continue and evolve in the next few years, with the development and improvement of some of the architectural and circuit strategies discussed in this paper, and very especially those based on the use of continuous-time circuits, enhanced by an intensive use of digital-assisted analog techniques, which will push $\Delta\Sigma$ converters forward.

Although lowpass $\Delta\Sigma$ ADCs will continue to dominate the marketplace, the push for improved efficiency and the direct digitization of RF signals will motivate increased commercialization of systems employing bandpass and quadrature bandpass ADCs—currently available in commercial form, but still relatively rare. More and more analog signal conditioning functions will be embedded in the feedback loop of the modulators, thus benefiting from noise-shaping to make the overall system more robust, compact and easy to adapt to different standards and operation modes. Some good examples of this trend are the digital-intensive $\Delta\Sigma$ receivers outlined in this work, but similar strategies could be also extended to other application scenarios such as wireless sensor networks and biomedical devices.

To fuel subsequent generations of $\Delta\Sigma$ ADCs, researchers will pursue more exotic implementations including time/frequency-interleaving, polyphase ADCs, ADCs with integrated sensors, photonic ADCs and more. These new inventions and design strategies will continue lifting up the energy efficiency and bandwidth of $\Delta\Sigma$ Ms to unprecedented levels, enhancing their dominant status over other types of ADCs, very especially in those applications requiring resolutions greater than 12 bits, while pushing their figures of merit towards physical limits, such as the room-temperature thermal noise limit (at low frequency ranges) and clock jitter (at high frequencies).

As a result of all these advances and design approaches, $\Delta\Sigma$ ADCs will gather more momentum for a fuller exploitation of the speed advantages of deep nanometer CMOS technologies, while getting along with their limitations for analog circuits. Addressing these issues in the next technology nodes will in turn result in more design innovations and new kinds of data converters. With so many obvious avenues going forward, the authors are confident that $\Delta\Sigma$ will reign supreme as the most varied, flexible and interesting ADC architecture for many years to come.

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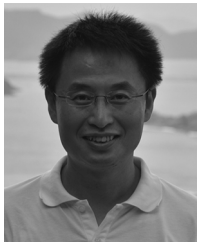
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